



# Cambridge (CIE) IGCSE Computer Science



Your notes

## Computer Sub-Systems

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Your notes

## Computer Sub-Systems

# Computer Sub-Systems

## What is a sub-system?

- A sub-system is a **smaller part of a computer system** that works together with other sub-systems to create fully functional computer system
- A car is a great example of using sub-systems
- A car will only function if its sub-systems **all work together**, they are:
  - **Engine**
  - **Breaks**
  - **Wheels etc.**
- Sub-systems can be **further broken down into even smaller sub-systems** such as:
  - **Engine** – spark plugs, sensors, pistons
- In computers, there are **five** main sub-systems:

| Central Processing Unit (CPU) | Memory                                                   | Storage                                         | Input devices                                        | Output devices                                                       |
|-------------------------------|----------------------------------------------------------|-------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------------|
| Executes instructions         | Stores data & instructions temporarily for the CPU (RAM) | Stores data and software permanently (HDD, SSD) | Allows a user to enter information (keyboard, mouse) | Displays information or creates a physical output (monitor, printer) |

- Computer sub-systems can be further broken down into smaller sub-systems such as:
  - **CPU** – control unit, registers & arithmetic logic unit (**ALU**)

## Advantages of sub-systems

- Can help **troubleshoot problems in a computer system**
- The ability to **isolate** a sub-system makes it **easier to identify and fix issues**, as each sub-system can be examined separately
- Developing software relies on the use of different sub-systems to ensure they **operate efficiently**

- Gives developers and designers **a clear picture** of how sub-systems help build complex systems



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## Problem Decomposition

# Problem Decomposition

## What is decomposition?

- Decomposition is the process of **breaking down a large problem** into a **set of smaller problems**
- Benefits of decomposition are:
  - **Smaller problems are easier to solve**
  - **Each smaller problem can be solved independently of the others**
  - **Smaller problems can be tested independently**
  - **Smaller problems can be combined to produce a solution to the full problem**
- Modern computer games are a good example of using decomposition to **break down the complexity** of the problem into more **manageable 'chunks'**
- Creating an entire game at once would be **challenging** and **inefficient**, so it could be decomposed into:
  - **Levels** - Levels can be designed/created/tested/ independently of other levels
  - **Characters** - The mechanics of characters in the game can be designed and created by a separate team
  - **Landscape** - The art team can work on the visual aspects of the game without needing to understand how the game is programmed
- Once all of the smaller problems are completed, joined together a complex game has been created

## Inputs, processes, outputs and storage

- Decomposing a problem requires developers to think about four component parts:
  - **Inputs**: data entered into the system
  - **Processes**: subroutines and algorithms that turn inputs and stored data into outputs
  - **Outputs**: data that is produced by the system, such as information on a screen or printed information
  - **Storage**: data that is stored on a physical device, such as on a hard drive or in memory whilst the program is running

## Example – Area of a rectangle program

|                  |                                                   |
|------------------|---------------------------------------------------|
| <b>Inputs</b>    | Width of the rectangle<br>Height of the rectangle |
| <b>Processes</b> | Width x height                                    |
| <b>Outputs</b>   | Calculated area of the rectangle                  |
| <b>Storage</b>   | Memory: width, height and area stored temporarily |



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